



Universitat Pompeu Fabra

Escola d'Enginyeria

Deep Learning

FINAL PROJECT

Emotion Recognition from Speech

CNN vs. LSTM: A Comparative and Explainability Study

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1 Introduction

Speech Emotion Recognition (SER) is the task of inferring a speaker’s affective state from audio signals. It has broad applications in areas such as human-computer interaction, mental health monitoring, and call-centre analytics. Despite decades of research, SER remains challenging due to high inter-speaker variability, class imbalance, and the inherent ambiguity of emotional expression. In this work, we evaluate a range of approaches on the RAVDESS dataset under a strict speaker-independent protocol, ensuring that no speaker seen during training or validation appears in the test set. This setup reflects real-world deployment conditions and provides a meaningful measure of generalisation. We compare a traditional SVM baseline and two deep learning architectures: a CNN trained on mel spectrograms, and an LSTM trained on MFCC sequences. As an additional experiment, we try the same classification problem with feature extraction using wav2vec 2.0. Models are evaluated using macro F1, per-class F1, accuracy, and confusion matrices. Our experiments show that fixed statistical features do not work across speakers, CNNs and LSTMs have trouble with few samples per class in this strict split, but wav2vec 2.0 pretrained representations close this gap by using sound knowledge from large unlabeled data.

2 State of the art

Early approaches to SER relied on hand-crafted acoustic features fed into traditional classifiers or shallow neural networks. Singh et al. Singh et al. 2021 combine manually extracted features with a hierarchical deep neural network operating on both audio and text, showing that multimodal inputs improve recognition. With the rise of deep learning, CNNs applied to spectrograms became a popular alternative: Kwon et al. Kwon et al. 2021 introduce a lightweight self-attention module on top of a CNN to focus on the most discriminative temporal regions, while Wijayasingha and Stankovic Wijayasingha and Stankovic 2021 demonstrate that CNNs with attention mechanisms are robust to noise when trained on spectrograms with augmentation. To better capture temporal dynamics, Atila and Şengür Atila and Şengür 2021 combine a 3D CNN with an LSTM and an attention mechanism, exploiting both spatial and sequential structure in the audio. More recently, self-supervised pretraining has emerged as a powerful alternative to learning from scratch: Pepino et al. Pepino, Riera, and Ferrer 2021 show that wav2vec 2.0 embeddings, passed through a simple dense layer or LSTM classifier, achieve strong SER performance with minimal task-specific training data, motivating our use of the same pretrained model as a transfer learning baseline.

3 Methodology

3.1 Data Analysis

We use the RAVDESS dataset from Livingstone and Russo 2019, which contains 1,440 audio-only speech recordings from 24 professional actors (12 male, 12 female) covering 8 emotion classes: neutral, calm, happy, sad, angry, fearful, disgust, and surprised. Each actor contributes 60 samples, resulting in a balanced class distribution across the dataset, which motivates the use of macro F1 as the primary evaluation metric.

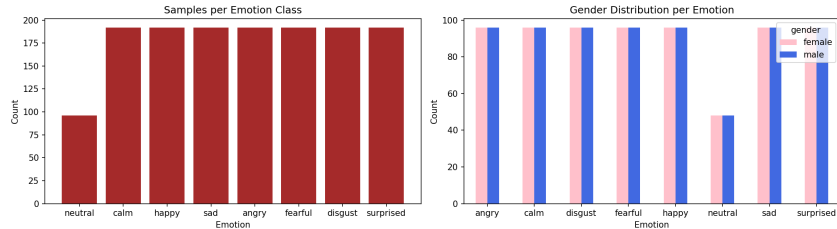


Figure 1: Emotion distribution per class and gender

Dataset split. To ensure that the model is evaluated on speakers it has never seen during training, we adopt a strict speaker-independent split: actors 1–18 are assigned to training (1080 samples, 75%), actors 19–21 to validation (180 samples, 12.5%), and actors 22–24 to test (180 samples, 12.5%). Actor assignment is mutually exclusive across splits by construction. Actors alternate by gender (odd = male, even = female).

Feature extraction. All audio files are resampled to 22050 Hz and padded or truncated to exactly 3 seconds (66150 samples). Two feature representations are extracted in parallel. For the CNN, we compute log-mel

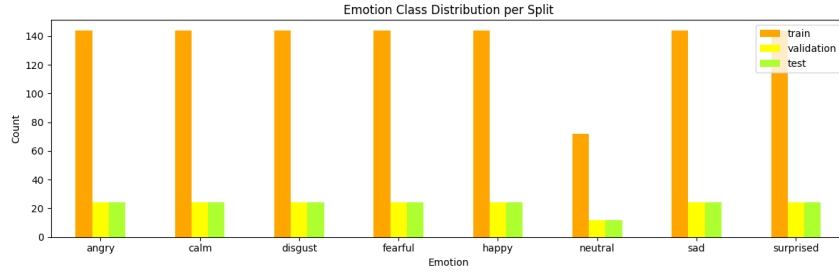


Figure 2: Emotion distribution per class and gender

spectrograms using a 2048-point FFT, a hop length of 512 samples, and 128 mel frequency bands, yielding a (128×130) matrix per clip, effectively a grayscale image of the sound. For the LSTM and SVM baseline, we extract 40 Mel Frequency Cepstral Coefficients (MFCCs) per frame, yielding a (40×130) sequence. MFCCs compress the spectral envelope into a compact representation well-suited for sequential modelling. All features are pre-extracted and saved as .npz files to avoid repeated computation across experiments.

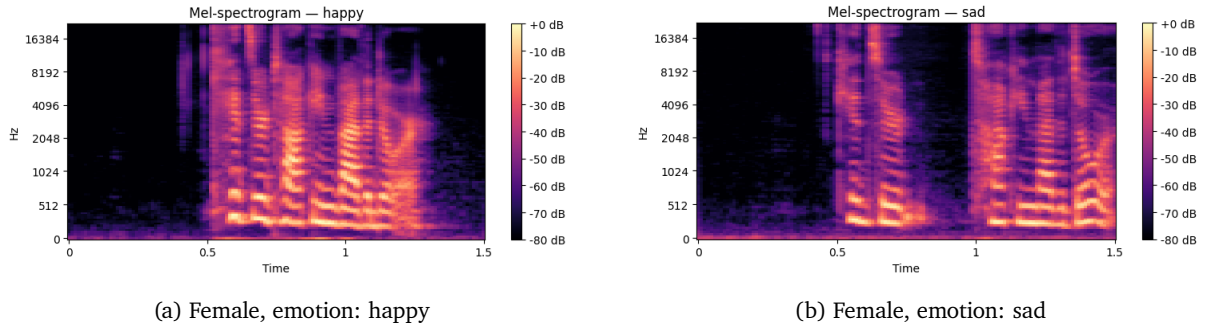


Figure 3: Spectrograms of the same sentence spoken by the same female speaker in two different emotions.

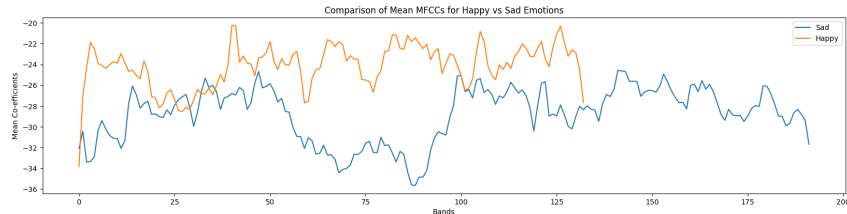


Figure 4: Comparison of two emotions' mean MFCC features over time.

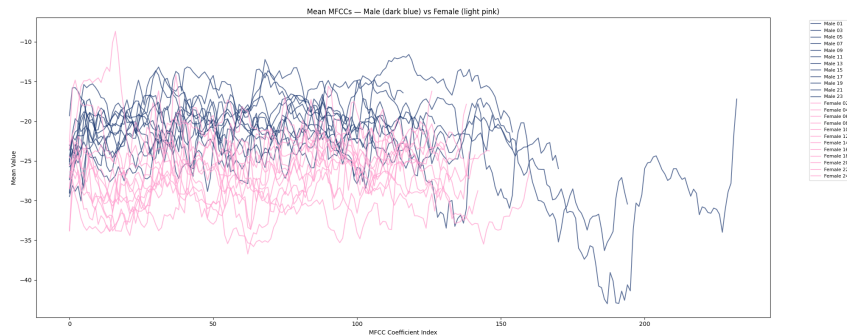


Figure 5: Mean MFCC features of all speakers in the train set saying the same sentence in the same emotion. Female speakers in pink; male speakers in navy.

From Figure 4 we observe visual differences in two samples that only differ in emotion. Implying that from

the input data, the models can learn to classify emotions. However, in Figure 5 we observe that the female audios have, in general, lower mean MFCC features. This shows that there is room for the model to learn to distinguish between people (for example gender-wise) rather than focusing on what different emotions look like. Therefore, we anticipate that the task will be difficult.

3.2 Models and Optimization

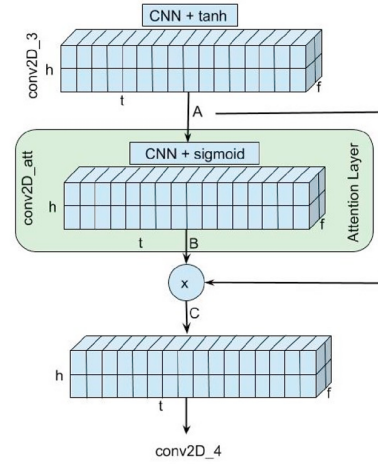
SVM Baseline. Each MFCC sequence of shape (40, 130) is reduced to a 120-dimensional vector by concatenating the per-coefficient mean, standard deviation, and average rate of change (delta) across time.

CNN Architecture. Because the Mel Spectrograms are just images, we can use them as input to a traditional 2D CNN. From the overview we obtained after reading the state of the art literature, we decided not to implement a CNN architecture from scratch. Having many CNN implementations that have already been optimized for the task is an advantage, and we can focus on the evaluation and comparison with the rest of models instead. The architecture we chose is from Wijayasingha and Stankovic 2021. It contains 4 convolutional layers with an additional attention mechanism in between the third and fourth layers. Figures 6a and 6b show the specific components and how they are layered. It receives the mel spectrograms as input with $(B, 1, 128, 130)$ dimensional vectors for a batch size of B .

FCNN architecture.

Layers
conv2D 32 3×3 1
Activation relu
Max Pooling 2×2 2
conv2D 64 3×3 1
Activation relu
Max Pooling 2×2 2
conv2D 16 3×3 1
Activation relu
Dropout 0.5
conv2D 7 3×3 1
Global Average Pooling
Activation softmax

(a) FCNN architecture



(b) Attention mechanism component with $B = \sigma(\text{conv2D}_{att}(A))$ and $C = A * B$.

Figure 6: CNN architecture proposed by Wijayasingha and Stankovic 2021.

LSTM Architecture. The motivation for using an LSTM is that emotion unfolds over time: a fearful voice doesn't reveal itself in a single frame but through how pitch, energy, and timbre evolve across the utterance. The LSTM operates on per-sample normalised MFCC sequences of shape (130, 40), where 130 is the number of time steps and 40 the feature dimension. Two models are evaluated: **1)** a standard 2-layer LSTM that extracts the last hidden state and passes it through a linear classifier, and **2)** a bidirectional 2-layer LSTM with dropout 0.3 that concatenates the final forward and backward hidden states before classification. Both are trained with `CrossEntropyLoss` and Adam ($\text{lr}=1\text{e-}3$, weight decay= $1\text{e-}4$) for 60 epochs, with time masking and frequency masking applied to training sequences as data augmentation.

wav2vec 2.0. As a transfer learning extension, we use facebook/wav2vec2-base, pretrained on 960 hours of unlabelled speech from LibriSpeech via self-supervised learning. All encoder weights are frozen, only a lightweight classifier is trained on top. Each 3-second audio clip is processed at 16 kHz, and the model's last hidden states are mean-pooled over time to produce a 768-dimensional embedding. The classifier consists of Linear(768-to-128)-to-ReLU-to-Dropout(0.2)-to-Linear(128-to-8). Two optimisers are compared: Adam ($\text{lr}=1\text{e-}2$, weight decay= $1\text{e-}4$) and SGD with momentum 0.9 ($\text{lr}=1\text{e-}3$, weight decay= $1\text{e-}6$). Both are trained for 100 epochs.

A note on comparing CNN and LSTM. The CNN gets mel spectrograms as input, while the LSTM gets MFCC sequences. So when we compare their performance, we are not just comparing architectures (convolutional

vs. recurrent) but two different ways of representing the audio. This is acceptable because both pipelines are standard in SER research, but the reader should keep in mind that we are comparing complete systems, not just the models themselves.

4 Experiments

4.1 Baseline

To train the SVM, because they are sensitive to feature magnitude, the features are standardised with zero mean and unit variance. The regularisation parameter C and kernel coefficient γ are selected via 5-fold cross-validated grid search on the training set.

4.2 CNN

The CNN training consists on three steps:

- a. *Baseline*: find approximately how many epochs the model needs to converge and the best learning rate and optimizer.
- b. *Ablation study*: independently remove **dropout**, the **attention mechanism**, and change the activation function of the third convolutional layer from Tanh to ReLU to see if the components are necessary in the architecture.
- c. *Improvement*: try adding BatchNorm, normalization, and augmentation to see if any of these components help improve performance.

For the baseline, we trained for 250 epochs to see how long the model needed to converge. Once established, we test for learning rate $1e-3$, $1e-4$, and $1e-5$, and compare the Adam optimizer (as used in Wijayasingha and Stankovic 2021 with Stochastic Gradient Descent with Momentum. Regarding normalization, we normalize per frequency-band, as time normalization does not make sense in this case because the speakers can start talking at different moments in time, and the sentences are different.

It is important to mention that this does not reproduce the steps in Wijayasingha and Stankovic 2021. First of all, their input images are different because they do not use the mel version of the spectrograms, and the augmentations they use are very specialized (they add noise via other audios and test the networks robustness). Our justification for using the mel spectrograms is that because they are normalized to human-distinguishable-frequencies, it would facilitate the network to distinguish emotions.

4.3 LSTM

The LSTM reads the MFCC sequence step by step and compresses it into a single hidden state that summarises the entire temporal trajectory, which is then passed to a linear classifier. Two configurations were evaluated. M1 is a standard 2-layer LSTM with hidden size 128, taking the last hidden state of the final layer as the sequence representation. M2 extends this by making the LSTM bidirectional, allowing it to read the sequence in both directions and concatenate the final forward and backward hidden states before classification, and adding dropout 0.3 between layers to regularise the larger parameter count. Both models are trained with CrossEntropyLoss and Adam ($lr=1e-3$, weight decay= $1e-4$) for 60 epochs. Since the training set is small, data augmentation is applied during training: random time masking zeros out a contiguous segment of up to 20 time steps, and frequency masking zeros out up to 10 consecutive MFCC coefficients, forcing the model to rely on partial information rather than memorising specific patterns.

4.4 wav2vec 2.0

Given the consistent underperformance of both the CNN and LSTM under the speaker-independent protocol, we hypothesised that the limiting factor was data scarcity rather than architectural choice. To test this, we replaced learned feature extraction with frozen representations from facebook/wav2vec2-base, a model pretrained on 960 hours of unlabelled speech. If the data hypothesis is correct, a simple classifier trained on top of these rich pretrained representations should substantially outperform models trained from scratch on the same 1080 samples. Two optimisers were compared for the lightweight classifier head: Adam ($lr=1e-2$, weight decay $1e-4$) and SGD with momentum 0.9 ($lr=1e-3$, weight decay $1e-6$), both trained for 100 epochs.

5 Results

5.1 Baseline

The best parameters for the SVC classifier from `sklearn` obtained using the `GridSearchCV` class are: `C=100` and `gamma='scale'`. The baseline training results are in Figure 7 and Table 1.

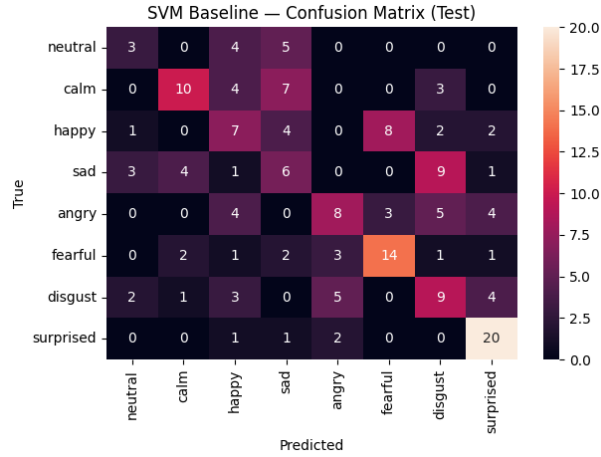


Figure 7: SVM Baseline Confusion Matrix on the test set.

General Metrics		Per class F1 score							
Accuracy	Macro F1	netural	calm	happy	sad	angry	fearful	disgust	surprised
0.428	0.414	0.29	0.49	0.29	0.24	0.38	0.57	0.34	0.71

Table 1: Baseline evaluation on test set.

5.2 CNN

The training curves of the original CNN's architecture in Figure 8 told us how many epochs the model needs to converge. We choose 200 as the validation loss stabilizes around that epoch, and training for longer leads to overfitting the network. This is the best performance with Adam optimizer and a learning rate of $1e-4$.

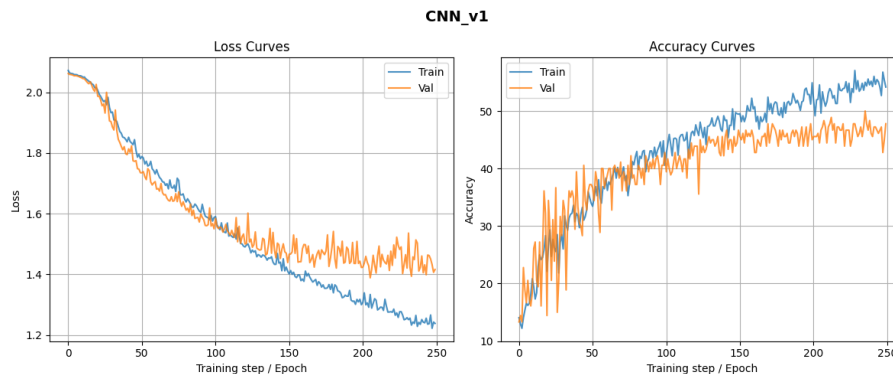
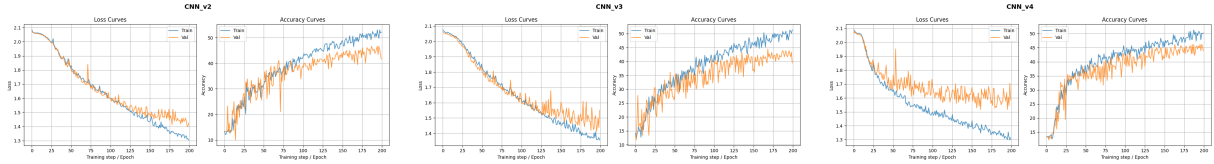


Figure 8: CNN training curves obtained from the Original Architecture in Wijayasingha and Stankovic 2021 with no changes. Peak accuracy: 51.1%

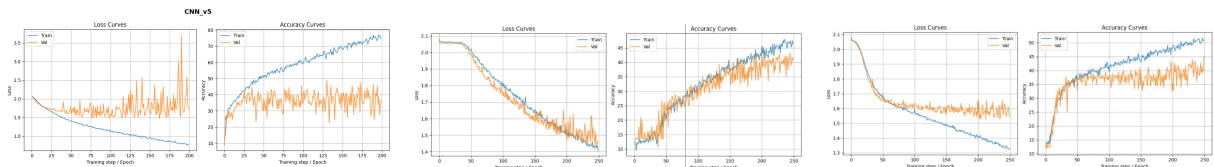
The results of the ablation study are shown in Figure 9. In the ablation study, we see that by removing dropout or the attention mechanism, and changing the activation function results in worst performance. Accuracy drops and we see how all three components are essential to the architecture to perform this task. Specially when removing the attention mechanism, because the model no longer has the layer that is helping find the parts in the image that distinguish the different patterns, it learns way less.



(a) CNN_v2: no dropout. Accuracy: 46%. Loss: 1,43. (b) CNN_v3: no attention mechanism. Accuracy: 42%. Loss: 1,44. (c) CNN_v4: ReLU instead of Tanh. Accuracy: 45%. Loss: 1,53.

Figure 9: Loss and accuracy curves of each training performed for the ablation study.

In Figure 10 we can see the result of adding the three mentioned components. By adding BatchNorm, gradients stabilize and the model trains faster at the cost of extreme overfitting. Because we do not have a lot of training data, we decided that trying to increase model depth to compensate the addition of BatchNorm was not worth it, and avoid overfitting with dropout is already included in the result. The normalization does not help the model learn better and we attribute this to the fact that the Mel spectrograms are already scaled to specific frequencies, by adding even more normalization, we are just losing information. Finally, the poor performance after adding augmentation in the train set is telling us that we made the task even harder for the model and the basic audio augmentations¹ we used do not help the model generalize and it even adds overfitting.



(a) CNN_v5: + BatchNorm. Accuracy: 48%. Loss: 1,51. (b) CNN_v6: + normalization. Accuracy: 44%. Loss: 1,43. (c) CNN_v7: + augmentation. Accuracy: 45%. Loss: 1,52.

Figure 10: Loss and accuracy curves of each training performed by adding the respective components.

To further investigate how the CNN is trained, we visualize the attention map of a specific sample, and the feature maps of the same example in the first three convolutional layers in Figures 11 and 12. In the activation map, the bright pixels coincide with the area where the voice frequency is plotted. And the feature maps show how the first convolutional layer is focusing on high-level-patterns, the second layer is starting to distinguish some structures, and the third layer has more abstract features. In these figures, we do not see any weird behaviour of the CNN.

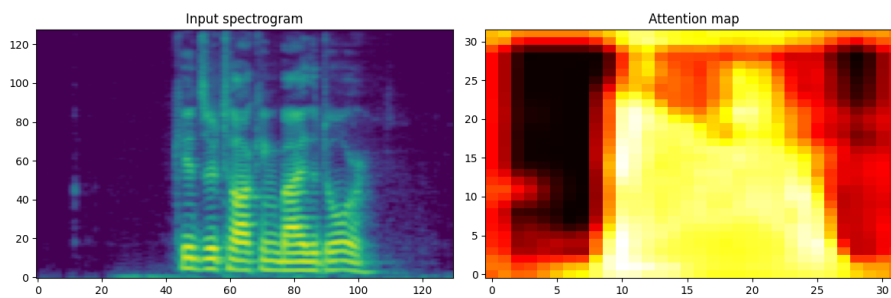


Figure 11: Attention map for a particular sample of the train set.

5.3 LSTM

M1, the standard 2-layer LSTM, reaches 40% validation accuracy within the first 30 epochs, but a growing gap between training and validation performance indicates the model is beginning to overfit (Figure 13a). Extending training to 60–100 epochs confirms this: validation accuracy flattens between 40–45% while training accuracy continues to rise (Figure 14). M2, the bidirectional variant with dropout 0.3, produces smoother

¹We used techniques like Gaussian noise addition, frequency band corruption, and time step corruption.

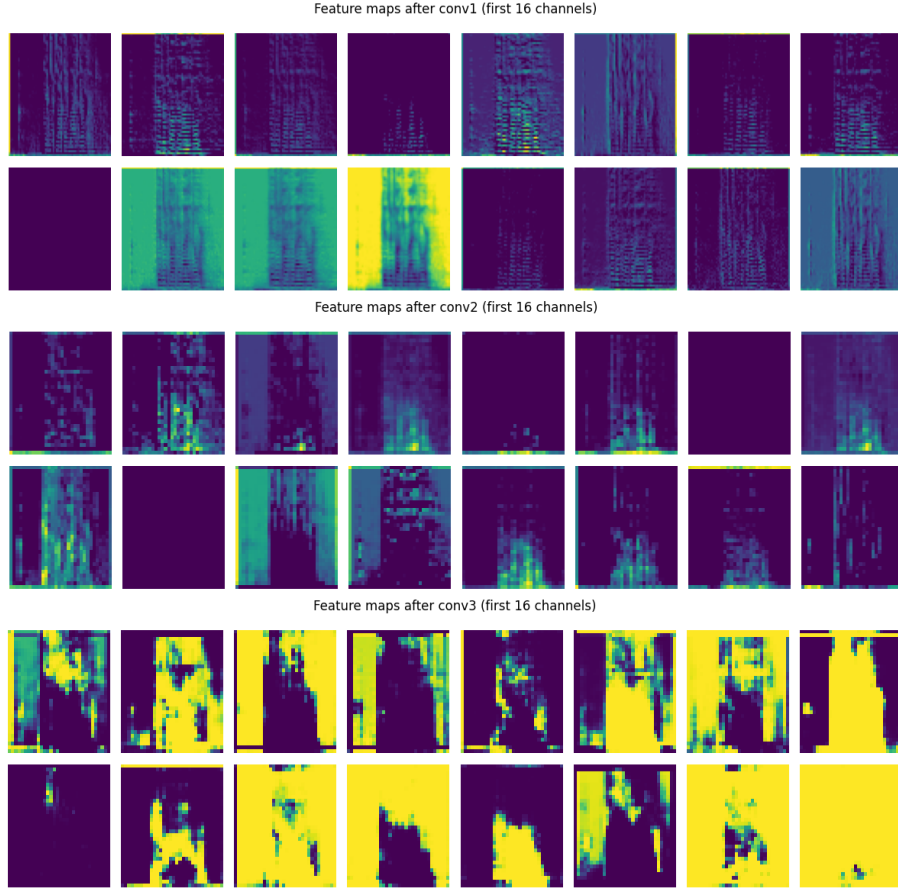


Figure 12: Feature maps for a specific input on the first three convolutional layers.

learning curves and a modest improvement to 44% validation accuracy with a smaller train/validation gap (Figure 13b), suggesting that bidirectionality and regularisation partially mitigate overfitting but do not resolve the underlying problem.

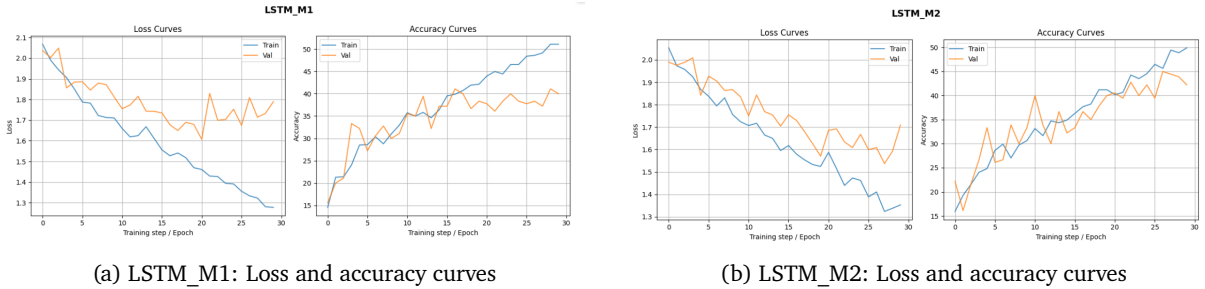


Figure 13: Loss and accuracy curves for both LSTM models, 30 epochs

The noisy, oscillating validation loss across both configurations points to a more fundamental issue: 180 validation samples are insufficient to provide a stable optimisation signal, and with only 140 training examples per class, the model struggles to learn speaker-invariant temporal patterns from scratch. M2 was selected for test evaluation on the basis of better generalisation.

5.4 wav2vec

Three optimizer configurations were compared during training (Figure 15). M2 (SGD, $lr=1e-3$) fails to learn effectively, with validation accuracy flattening around 35–38% across 100 epochs. Train and validation curves move nearly in lockstep, suggesting the learning rate is too low for the shallow classifier to converge meaningfully on top of frozen features. M1 (Adam, $lr=1e-2$) and M2.1 (SGD, $lr=1e-2$) both learn substantially faster

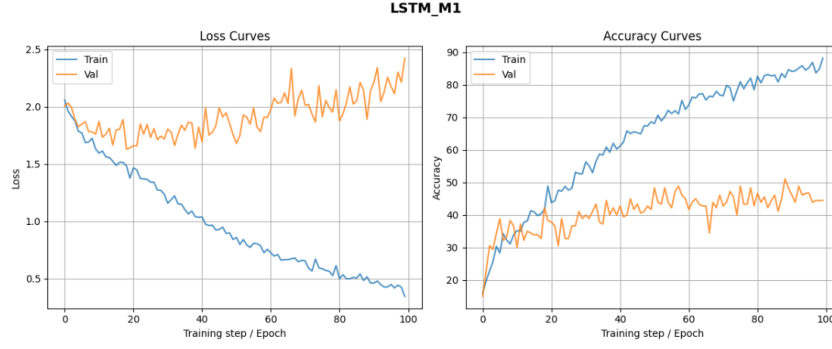


Figure 14: LSTM_M1: Loss and accuracy curves, 100 epochs.

and reach higher validation accuracy (55–60%), but M1 shows a noisier validation curve while M2.1 converges more smoothly, with validation accuracy steadily climbing to 58% before stabilising. M2.1 was selected for test evaluation.

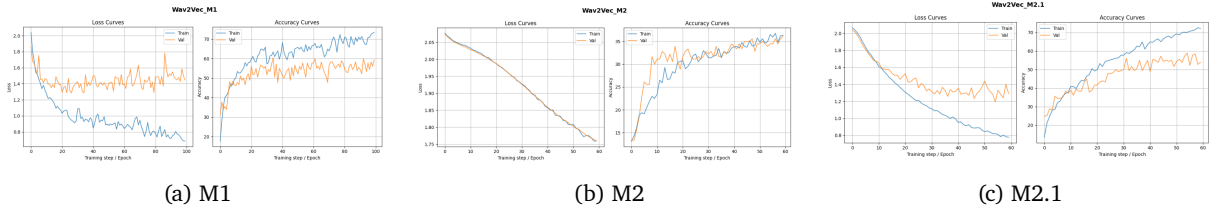


Figure 15: Loss and accuracy curves for wav2vec models

5.5 Models Comparison

After the training of the three models, we can compare the results evaluated on the test set. The results are shown in Figure 16 and Table 2.

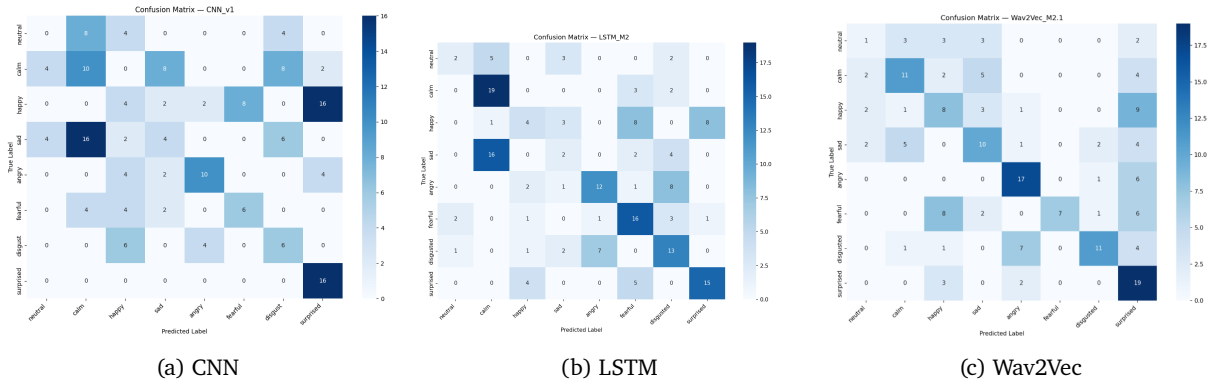


Figure 16: Confusion matrices of the three models on the test set.

In Figure 17 shows a visual representation of the per-class F1 scores.

Regarding the per-class F1 score, we can see some patterns. The *surprised* and *calm* classes are consistently the easiest to classify, with most models achieving F1 scores above 0.45. This suggests that these emotions have more distinctive and speaker-invariant characteristics. In contrast, *neutral* is the hardest class across models, with some scoring even 0.0. This is likely because it can easily be confused with other low-arousal emotions, such as calm or sad. The *angry* class benefits the most from richer representations: wav2vec achieves 0.67, the highest per-class score in the table. The *happy* and *disgust* classes remain difficult for the CNN and LSTM, yet the SVM and wav2vec recover meaningfully. This suggests that these emotions require either compact statistical summaries (SVM) or rich pretrained representations (wav2vec) to generalise across speakers.

Table 2: Metrics results on the test set

	General Metrics		Per class F1 score							
	Accuracy	Macro F1	neutral	calm	happy	sad	angry	fearful	disgust	surprised
CNN_v1	31.11	30.46	0.00	0.29	0.14	0.16	0.56	0.40	0.30	0.59
LSTM_M1	42.22	36.40	0.00	0.75	0.15	0.33	0.37	0.48	0.23	0.58
Wav2Vec_M2.1	48.33	46.05	0.21	0.49	0.36	0.43	0.67	0.40	0.62	0.51
SVM (baseline)	42.80	41.14	0.45	0.69	0.42	0.46	0.43	0.38	0.38	0.47

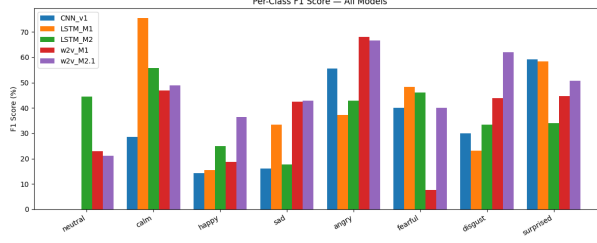


Figure 17: F1 per class.

5.6 Attention Map Analysis for Happy-Surprised Confusion

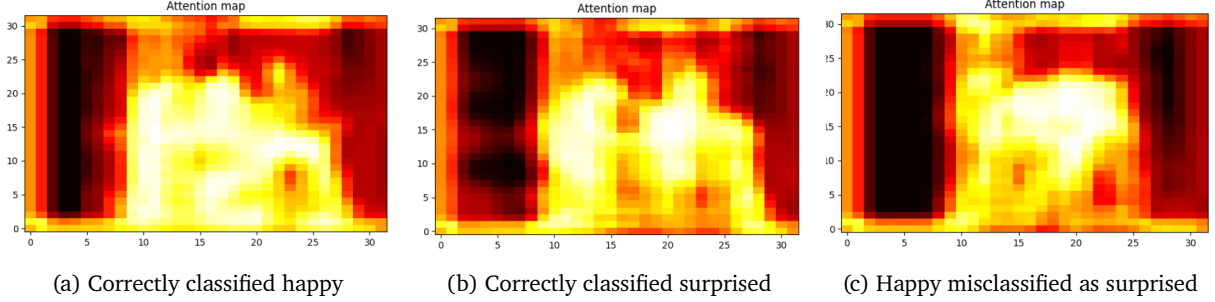


Figure 18: Attention maps for happy and surprised test samples.

Figure 18 shows attention maps for three test samples. In the correctly classified happy (subfigure 18a), attention is concentrated on a compact mid-frequency band (x-axis: 10–25, y-axis: 0–20) across most time steps. In the correctly classified surprised (subfigure 18b), attention is also focused, but the dark (low-activation) area is larger than in happy. The x-axis remains 10–25 and the y-axis 0–20, but the pattern appears more intermittent, reflecting the sudden pitch jumps typical of surprise. The misclassified happy sample (subfigure 18c) shows an attention pattern that does not clearly resemble either of the correct ones, and the dark area is larger and more similar to surprised than to happy: the activation is shifted upward (y-axis: 0–25) and spread across a wider band (x-axis: 10–25), losing the compact mid-frequency focus. We hypothesise that this particular happy sentence had a slightly higher or more variable pitch than average, causing the CNN’s attention mechanism to misinterpret it as surprise. This explains why happy ($F1=0.14$) and surprised ($F1=0.59$) are confused in Table 2, when the spectral pattern of happy deviates from the prototype, it falls into the surprised region.

6 Discussion

CNN performance over literature. Although we use the exact same architecture as in the literature for the CNN, our results are way less promising. We attribute this gap to two key differences. First, our input representation differs: we use log-mel spectrograms rather than the raw spectrograms used in the reference, and although mel-scaling was motivated by its alignment with human auditory perception, it may discard information. Second, we enforce speaker-independent split, while most references do not explicitly report such protocol, or they even simplify the classification task by merging similar emotion classes into fewer categories. Both factors make our experimental setup considerably harder.

LSTM over CNN. Our results confirm that the LSTM generalises better than the CNN under the speaker-independent protocol, despite operating on a lower-dimensional input. We attribute this to a combination of

representational and architectural factors. Mel spectrograms are high-dimensional (128×130) and preserve fine-grained spectral detail that a CNN can exploit to fit speaker-specific patterns, even with dropout and the attention mechanism in place. MFCCs, by contrast, are a compact representation that suppresses speaker identity cues and retains the temporal envelope of speech (maybe a more emotion-relevant signal). The LSTM, however, is able to observe how the sequence changes over time and discover some patterns related to how different emotions change. The CNN, treating the spectrogram as a static image, is more sensitive to spectral shape than to temporal evolution, which under a small speaker-independent split translates directly into poorer generalisation.

Wav2Vec dominance. The strong performance of wav2vec 2.0 relative to both the CNN and LSTM, despite using only a frozen encoder and a lightweight two-layer classifier, confirms our hypothesis that data scarcity is the primary bottleneck of this study. Pretrained on 960 hours of unlabelled speech, the wav2vec encoder produces 768-dimensional embeddings that capture rich, speaker-invariant acoustic representations which our models trained from scratch on 1,080 samples cannot replicate. The consistent improvement across nearly all emotion classes, most notably on *disgust* (0.62) and *angry* (0.67), suggests that the pretrained representations encode better information for emotion discrimination, without needing to be fine-tuned. This result underlines a fundamental limitation of training deep models from scratch on small, constrained datasets, and motivates the use of transfer learning as a default strategy for low-resource SER tasks.

Baseline over CNN and LSTM. A surprising result is that the SVM baseline outperforms both the CNN and the LSTM in macro F1 (41.14 vs. 30.46 and 36.40, respectively), despite using the simplest input features. The most likely explanation is that the CNN and LSTM have far more parameters than the SVM, and with only 1,080 training samples they simply do not have enough data to learn useful patterns without overfitting to speaker-specific details. The SVM, working in a much smaller and more compact feature space, is less affected by this problem and can find a reasonable decision boundary with the data available. This confirms that in small dataset settings, simpler models can outperform more complex ones.

Explainability and limitations. Our attention map analysis (Section 5.6) shows why the CNN confuses happy with surprised: pitch deviation shifts attention toward the surprised pattern. A similar analysis for the LSTM was not possible given time and space, but future work should compare both models internal representations, examining temporal attention or hidden state trajectories to explain why the LSTM generalises better.

7 Conclusions

The main takeaway of this work is that **good input features matter more than complex models when the dataset is small**. The SVM, even though it only used a 120-dimensional summary of MFCCs, performed better than both the CNN and the LSTM. Wav2vec 2.0 did even better, despite using the simplest classifier. This pattern shows that under a strict speaker-independent setup with only 1,080 training samples, neither the CNN nor the LSTM has enough data to learn stable, speaker-invariant features. Because of this, future work should focus on collecting more speakers or using augmentations that increase speaker diversity, rather than trying new architectures.

The LSTM outperforming the CNN is also easy to explain. Emotion is a *temporal* signal: it depends on how pitch, energy, and rhythm change over time, not on single frames. The LSTM is built to model sequences, while the CNN treats the spectrogram like an image, which makes it more sensitive to speaker-specific patterns. Under a strict speaker-independent split, this becomes a problem: any model that captures speaker identity instead of emotion will not generalise. Our results therefore support using sequence models over image-based ones for SER when data is limited and speakers vary a lot.

Finally, this work shows the cost of using a rigorous evaluation protocol. Our strict split—where no speaker appears in more than one partition—matches real deployment but makes the task much harder than the random splits often used in the literature. The gap between our CNN results and those in Wijayasingha and Stankovic 2021 is likely due to this difference in evaluation, not because our model is worse. As a field, SER would benefit from standardising speaker-independent evaluation and reporting per-speaker generalisation, so results reflect real robustness instead of within-speaker memorisation.

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